7 Stokes' Theorem and the Divergence Theorem

7.1 Stokes' Theorem

Definition 7.1.1. Let σ be an oriented surface with unit normal field $\mathbf{n}$, and let C be the boundary curve of σ. The **positive orientation** of C is the direction determined by the chosen orientation of σ. Equivalently, if one walks along C with the head pointing in the direction of $\mathbf{n}$, then the surface remains on the left.

Theorem 7.1.1 (Stokes' Theorem)**.** *Let σ be a piecewise smooth oriented surface bounded by a positively oriented, piecewise smooth, simple closed curve C. Suppose that*

$$\mathbf{F}(x,y,z) = f(x,y,z)\hat{\mathbf{i}} + g(x,y,z)\hat{\mathbf{j}} + h(x,y,z)\hat{\mathbf{k}}$$

is a vector field whose component functions have continuous first-order partial derivatives on an open set containing σ. If $\mathbf{n}$ is the chosen unit normal field for σ, then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_\sigma (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n}\, dS.$$

Proof: The proof is omitted.

Example 1. Evaluate

$$\oint_C \left(\frac{1}{2} y^2\, dx + z\, dy + x\, dz \right),$$

where C is the curve of intersection of the plane $x + z = 1$ and the ellipsoid

$$x^2 + 2y^2 + z^2 = 1,$$

with orientation counterclockwise as viewed from above.

Example 2. Evaluate

$$\iint_\sigma (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n}\, dS,$$

where

$$\mathbf{F}(x,y,z) = x\hat{\mathbf{i}} + y^2\hat{\mathbf{j}} + ze^{xy}\hat{\mathbf{k}}$$

and σ is the part of the surface

$$z = 1 - x^2 - 2y^2, \qquad z \geq 0,$$

with upward orientation.

Remark. Green's Theorem can be viewed as a special case of Stokes' Theorem. Indeed, if D is a region in the xy-plane, $\sigma = D$, $\mathbf{n} = \hat{\mathbf{k}}$, and

$$\mathbf{F}(x,y,z) = M(x,y)\hat{\mathbf{i}} + N(x,y)\hat{\mathbf{j}},$$

then

$$\operatorname{curl} \mathbf{F} = \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \hat{\mathbf{k}},$$

and Stokes' Theorem gives

$$\oint_C M\, dx + N\, dy = \iint_D \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA.$$

7.2 The Divergence Theorem

Definition 7.2.1. A surface σ is called a **closed surface** if it is the boundary of a solid region in $\mathbb{R}^3$. For a closed surface, the **positive orientation** is the outward orientation.

Theorem 7.2.1 (Divergence Theorem)**.** *Let σ be a closed, piecewise smooth, orientable surface that encloses a solid region R in $\mathbb{R}^3$. Suppose that $\mathbf{F}$ is a vector field whose component functions have continuous first-order partial derivatives on an open set containing R and its boundary σ. If $\mathbf{n}$ is the outward unit normal field for σ, then*

$$\iint_\sigma \mathbf{F}\cdot\mathbf{n}\,dS = \iiint_R \operatorname{div}\mathbf{F}\,dV.$$

Proof: The proof is omitted.

Example 3. Evaluate

$$\iint_\sigma \mathbf{F}\cdot\mathbf{n}\,dS,$$

where

$$\mathbf{F}(x,y,z) = x^2\hat{\mathbf{i}} + xy\hat{\mathbf{j}} + x^3y^3\hat{\mathbf{k}}$$

and σ is the surface of the tetrahedron bounded by the plane

$$x+y+z=1$$

and the coordinate planes, with outward unit normal vector $\mathbf{n}$.

Example 4. Evaluate

$$\iint_\sigma \mathbf{F}\cdot\mathbf{n}\,dS,$$

where

$$\mathbf{F}(x,y,z) = xy\hat{\mathbf{i}} - z^2\hat{\mathbf{k}}$$

and σ is the union of the upper five faces of the unit cube $[0,1]^3$, oriented outward from the cube. Since σ is not a closed surface, close it by adding the bottom face and then use the Divergence Theorem.

Divergence Viewed as Flux Density

Suppose that G is a small solid region centered at a point P_0, and suppose that its boundary surface, denoted by $\sigma(G)$, is oriented outward. Let $\mathbf{F}$ be a vector field whose component functions have continuous first-order partial derivatives in an open set that contains G. Denote the volume of G by $\operatorname{Vol}(G)$ and the flux of $\mathbf{F}$ across $\sigma(G)$ by $\Phi(G)$. If $\operatorname{div}\mathbf{F}$ is continuous on G, then the value of $\operatorname{div}\mathbf{F}$ will not vary much from its value at the center, $\operatorname{div}\mathbf{F}(P_0)$, when G is small. Thus, we may approximate $\operatorname{div}\mathbf{F}$ by the constant $\operatorname{div}\mathbf{F}(P_0)$ on G. Therefore,

$$\begin{aligned}
\Phi(G) &= \iint_{\sigma(G)} \mathbf{F}\cdot\mathbf{n}\,dS \\
&= \iiint_G \operatorname{div}\mathbf{F}\,dV \\
&\approx \operatorname{div}\mathbf{F}(P_0)\iiint_G dV \\
&= \operatorname{div}\mathbf{F}(P_0)\operatorname{Vol}(G).
\end{aligned}$$

This implies

$$\operatorname{div}\mathbf{F}(P_0) \approx \frac{\Phi(G)}{\operatorname{Vol}(G)}.$$

As the region G shrinks to the point P_0, this becomes

$$\operatorname{div}\mathbf{F}(P_0) = \lim_{G\to P_0} \frac{1}{\operatorname{Vol}(G)} \iint_{\sigma(G)} \mathbf{F}\cdot\mathbf{n}\; dS,$$

where $G \to P_0$ means that G shrinks to P_0 in all directions, for example $\operatorname{diam}(G) \to 0$. Thus, if $\mathbf{F}$ denotes the velocity field of a fluid in steady-state flow, then $\operatorname{div}\mathbf{F}$ can be interpreted as the limiting outward flux per unit volume at a point.

Exercises for Lecture 7

1. Let σ be an oriented surface with boundary curve C. Explain what is meant by the positive orientation of C induced by the orientation of σ.

2. Use Stokes' Theorem to evaluate

$$\oint_C \mathbf{F}\cdot d\mathbf{r},$$

where

$$\mathbf{F}(x,y,z) = y\hat{\mathbf{i}} + z\hat{\mathbf{j}} + x\hat{\mathbf{k}}$$

and C is the boundary of the triangle in the plane

$$x + y + z = 1$$

in the first octant. Use the orientation corresponding to the normal vector with positive z-component.

3. Use Stokes' Theorem to evaluate

$$\oint_C \mathbf{F}\cdot d\mathbf{r},$$

where

$$\mathbf{F}(x,y,z) = -z\hat{\mathbf{i}} + x\hat{\mathbf{j}} + y\hat{\mathbf{k}}$$

and C is the circle

$$x^2 + y^2 = 4, \qquad z = 3,$$

oriented counterclockwise as viewed from above.

4. Let

$$\mathbf{F}(x,y,z) = x^2\hat{\mathbf{i}} + xy\hat{\mathbf{j}} + z\hat{\mathbf{k}}.$$

Use Stokes' Theorem to evaluate

$$\iint_\sigma (\operatorname{curl}\mathbf{F})\cdot\mathbf{n}\; dS,$$

where σ is the upper hemisphere

$$x^2 + y^2 + z^2 = 9, \qquad z \geq 0,$$

with outward orientation.

5. Use the Divergence Theorem to evaluate the outward flux

$$\iint_\sigma \mathbf{F} \cdot \mathbf{n}\, dS,$$

where

$$\mathbf{F}(x, y, z) = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

and σ is the sphere

$$x^2 + y^2 + z^2 = a^2, \qquad a > 0.$$

6. Use the Divergence Theorem to evaluate the outward flux of

$$\mathbf{F}(x, y, z) = x^2\hat{\mathbf{i}} + y^2\hat{\mathbf{j}} + z^2\hat{\mathbf{k}}$$

through the boundary of the cube

$$0 \le x \le 1, \qquad 0 \le y \le 1, \qquad 0 \le z \le 1.$$

7. Use the Divergence Theorem to find the outward flux of

$$\mathbf{F}(x, y, z) = xy\hat{\mathbf{i}} + yz\hat{\mathbf{j}} + zx\hat{\mathbf{k}}$$

through the surface of the tetrahedron bounded by the coordinate planes and the plane

$$x + y + z = 2.$$

8. Determine whether the Divergence Theorem can be applied directly to each of the following surfaces. Give a reason for your answer.

   (a). The surface of a sphere.

   (b). The upper hemisphere only.

   (c). The boundary of a rectangular box.

   (d). The lateral surface of a cylinder without its top and bottom disks.